

Java

1. Basic reflection

Procedure 1.1 (Enumerating methods of class)

```
Class clazz = String.class; // by class name
Class clazz = instance.getClass(); // get
from an instance

for(java.lang.reflect.Method m :
clazz.getMethods()) {
    System.out.println(m);
}
```

Procedure 1.2 (Finding fully-qualified class name)

```
Class clazz = String.class; // by class name
Class clazz = instance.getClass(); // get
from an instance

System.out.println(clazz.getName());
System.out.println(clazz.getCanonicalName());
```

2. Common packages

- java.util — collections (List, Map, Set, ArrayList, HashMap, Scanner, Random, etc.)
- java.io — file/stream I/O (File, BufferedReader, IOException)
- java.lang — core classes, **auto-imported already** (String, Object, Math, Integer, etc. — no import needed)
- java.lang.reflect — reflection (Method, Field, Modifier)
- java.time — dates/times (LocalDate, LocalDateTime, Duration)
- java.util.stream — streams (Stream, Collectors)
- java.util.concurrent — threading utilities

3. Footguns

- == tests pointer equality, not value equality, if the arguments are objects
 - Use equals for value equality on common classes
- Similarly, don't use <, >, <=, >= on objects
 - Use compareTo, which returns positive value if LHS > RHS, negative value if LHS < RHS, and 0 if equal